

Trainings ▾

Initial Check

Misfire Using Heat Indicator Marker

Note : This procedure may **not** be effective when the symptoms occur **only** at low idle or when the engine is cold.

⚠ WARNING ⚠

The engine temperature, when performing in-chassis exhaust temperature testing, can be excessive. To reduce the possibility of serious injury wait until the engine coolant temperature is below 50°C [122°F].

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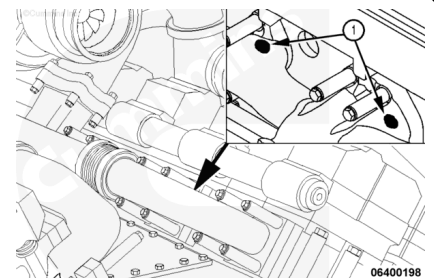
If the engine exhibits symptoms of a possible injector **not** firing, perform the following check:

Operate the engine to normal operating coolant temperature of 80°C [180°F].

Stop the engine and allow the exhaust manifold to cool for 10 minutes.

Apply a mark (1) using a 204°C [400°F] temperature indicator marker, (Part Number 3165163), on the surface of all exhaust manifold ports. The mark should be as close to the cylinder head as possible, but not directly on the exhaust manifold mounting flange.

Start the engine and operate under light load for a short time. Do not operate the engine under heavy load or for an extended period of time as this may cause inaccurate results.

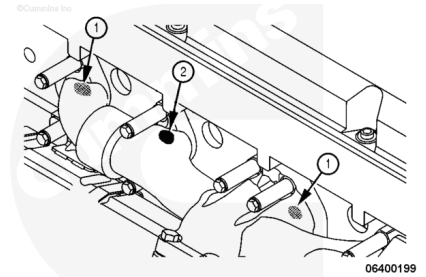


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If the 204°C [400°F] mark melts (1) the color will disappear and leave **only** a transparent glazed appearance where the mark was. This indicates that the cylinder is operating to a temperature above 204°C [400°F] .

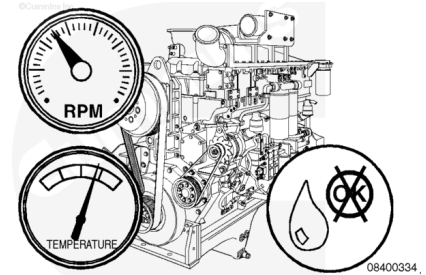
If the 204°C [400°F] mark does **not** melt (2) the mark color remains present. This indicates that the cylinder is **not** operating to temperature of 204°C [400°F], indicating a possible injector malfunction. Refer to the trouble shooting tree t062 for additional troubleshooting information.

Note : It is **not** recommended to use the temperature indicator markers to troubleshoot for cylinders that are operating too hot.

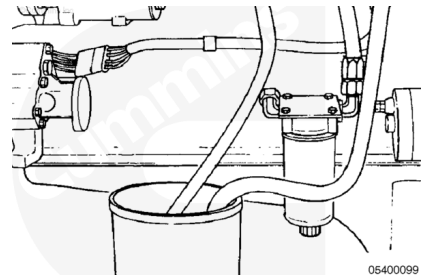


Flourescent Tracer Dye Test

Run the engine until the water temperature is above 80°C [180°F].



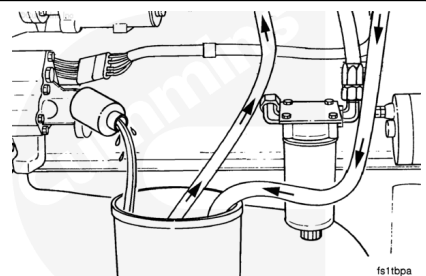
Remove the fuel pump return and supply lines and install a 50 litre [15 gallon] isolated fuel supply tank.



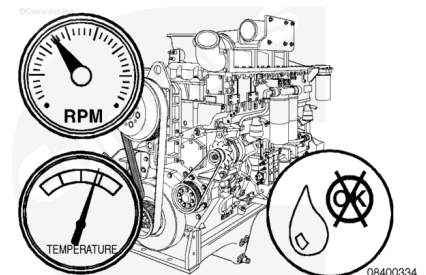
Add one unit of fluorescent tracer, Part Number 3376891, to each 38 litres [10 gallons] of fuel.

Preheat the black light.

Remove the rocker lever cover mounting hardware, but do **not** remove the rocker lever covers.



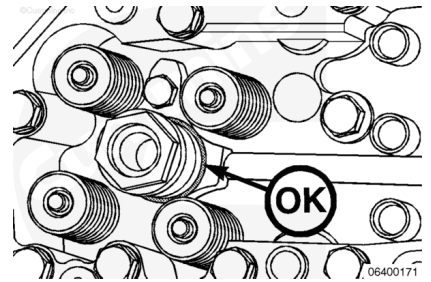
Idle the engine for 5 to 10 minutes or until normal operating temperature is reached to allow the dye to circulate through the fuel system.



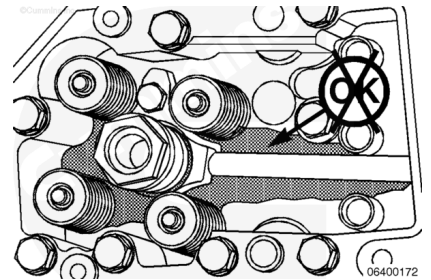
Shut the engine off.

Immediately remove the rocker lever covers and inspect each cylinder head using the black light.

Some leakage around the injector is normal.



Replace the injector or injector o-rings if an abnormal amount of leakage is detected.



Preparatory Steps

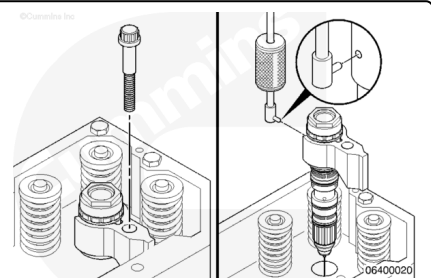
- Remove the rocker lever assembly. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/-003-009.html>).

Remove

Remove the injector hold-down capscrew. Remove the injector and clamp.

Use injector removal/installation tool, Part Number 3824830, to remove the injectors. Insert the pin of the tool in the hole provided in the body of the injector. The hole faces the front of the engine.

If the injector removal/installation tool mentioned above is **not** available, carefully use a pry bar. Pry upward on the injector against the cylinder head.



Inspect for Reuse

Remove the sealing ring (1) and note the cylinder location.

Sealing rings are available in different thickness sizes to adjust the injector protrusion.

The four new o-rings **must** be installed and oriented correctly on the injector. Identify the o-rings to install them in the correct grooves.

Remove the o-rings (2, 3, 4, and 5).

Use a lint-free cloth to clean the exterior of the injector.

Carefully check the area where the sealing ring touches the injector.

Inspect the o-ring grooves for damage.

Inspect the injector body and cup retainer for cracks or other damage.

Inspect the injector links for damage, excessive wear, and pitting or scoring on the ball ends.

If the link is damaged or pitting/scoring can be seen or felt, the link **must** be replaced.

O-ring (2) is brown and is smaller in diameter than the other three.

O-ring (3) is black.

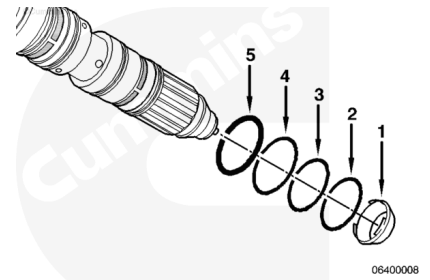
O-ring (4) is brown and is larger in diameter than (2), but has a smaller cross section than o-ring (5).

O-ring (5) is brown and is the largest in diameter and cross section.

Do **not** lubricate the o-rings until the injector is ready for installation in the cylinder head.

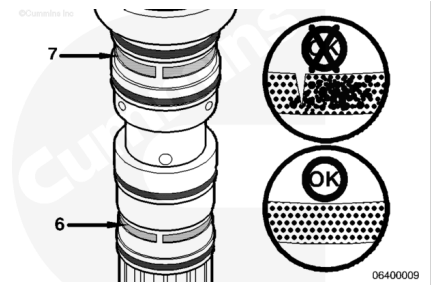
Install four new o-rings on the injector. Do **not** twist the o-rings.

Install the proper size sealing ring (1).



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

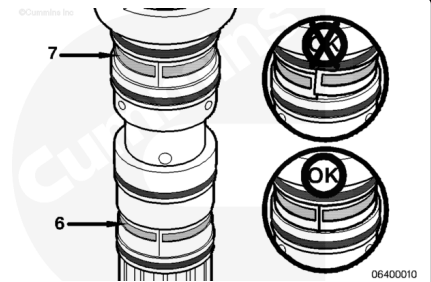


Use solvent when cleaning the filters. The screens **must** be thoroughly dry prior to installation.

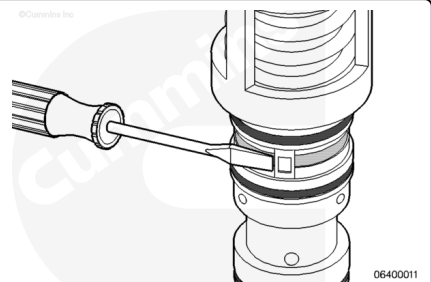
Check the rail (6) and timing (7) filter screens for debris, tears, or punctures. Replace the filter screens if damaged or dirty.

Check the rail (6) and timing (7) filter for proper installation.

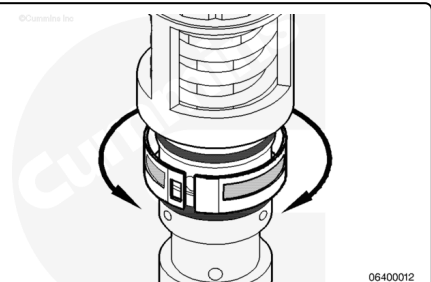
Screens **must** be securely snapped into the correct position.



Use a small screwdriver to remove the filter screen by gently prying up on the connector clip.

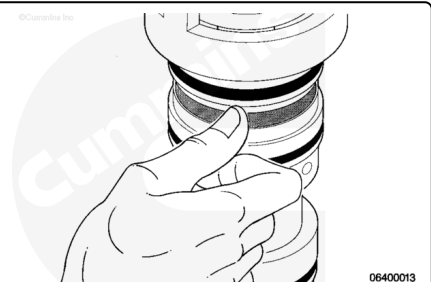


Wrap the filter screen around the inlet groove.



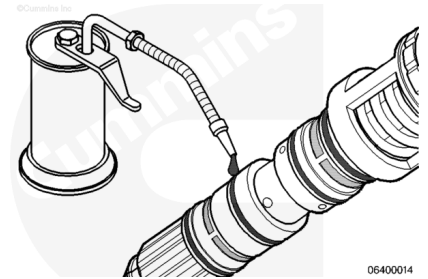
Apply pressure on the filter screen connector clip.

A properly secured connector will create a single, snapping sound.

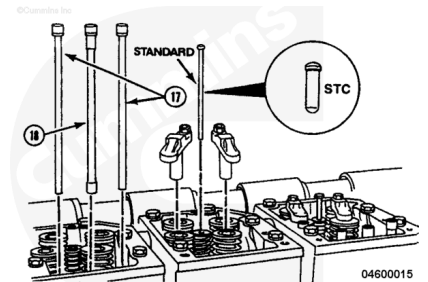


Install

Use clean 15W-40 engine oil to lubricate the o-rings.



Install the hold-down clamp on the side of the injector.

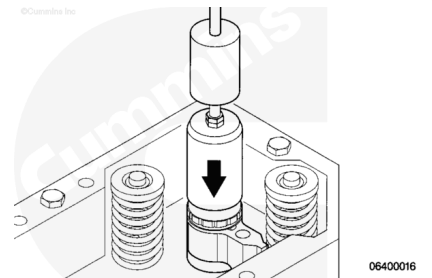


Install the injector and hold-down clamp in the cylinder head.

Use the injector removal/installation tool, Part Number 3824830, to set the injector in the bore.

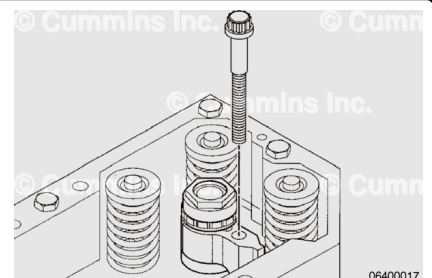
The slide hammer will make a dull sound when the injector is seated properly.

Note : If the injector removal/installation tool, Part Number 3824830, is not available, use a 40 mm [1 9/16 in] socket with an extension and a rubber mallet to install the injector.



Use clean 15W-40 engine oil to lubricate the injector hold-down capscrew.

Install the injector hold-down capscrew.



Torque Value:	Step 1	34 n.m [25 ft-lb]
	Step 2	Tighten 90°

Finishing Steps

- Install the rocker lever assembly. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures//-003-009.html>).
- Adjust the valves and injectors. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures//-003-006.html>).
- Install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures//-003-011.html>).

